```
import pandas as pd
import numpy as np

import re
import string
from nltk.corpus import stopwords
from nltk.stem import PorterStemmer

from sklearn.feature_extraction.text import CountVectorizer, TfidfVectorizer

from sklearn.model_selection import train_test_split
from sklearn.naive_bayes import MultinomialNB

from sklearn.metrics import accuracy_score, precision_score, recall_score, f1_score, confusion_mat
import seaborn as sns
import matplotlib.pyplot as plt
```

```
# Load dataset
df = pd.read_csv("/content/news.csv", encoding="latin-1")

# Display the actual column names
print("Actual column names:", df.columns.tolist())
```

```
  label                                            message
0  FAKE  Daniel Greenfield, a Shillman Journalism Fello...
1  FAKE  Google Pinterest Digg Linkedin Reddit Stumbleu...
2  REAL  U.S. Secretary of State John F. Kerry said Mon...
3  FAKE  â Kaydee King (@KaydeeKing) November 9, 2016...
4  REAL  It's primary day in New York and front-runners...
Dataset size: (6335, 2)
Class distribution:
 label
REAL    3171
FAKE    3164
Name: count, dtype: int64
```

```
import nltk
nltk.download('stopwords')

stop_words = set(stopwords.words('english'))
stemmer = PorterStemmer()

def preprocess_text(text):
    text = text.lower()
    text = re.sub(r'\d+', '', text)
    text = text.translate(str.maketrans('', '', string.punctuation))
    words = text.split()
    words = [stemmer.stem(w) for w in words if w not in stop_words]
    return " ".join(words)

df['clean_message'] = df['message'].apply(preprocess_text)
```

```
[nltk_data] Downloading package stopwords to /root/nltk_data...
[nltk_data]   Unzipping corpora/stopwords.zip.
```

```
vectorizer = TfidfVectorizer(max_features=3000)
X = vectorizer.fit_transform(df['clean_message'])
```

```
y = df['label'].map({'ham':0, 'spam':1})

print("Feature matrix shape:", X.shape)
print("Sample feature names:", vectorizer.get_feature_names_out()[:20])
```

```
Feature matrix shape: (6335, 3000)
Sample feature names: ['abandon' 'abc' 'abedin' 'abil' 'abl' 'abort' 'abroad' 'absenc' 'absolut'
 'abus' 'academ' 'accept' 'access' 'accompani' 'accomplish' 'accord'
 'account' 'accur' 'accus' 'achiev']
```

```
y = df['label'].map({'FAKE':0, 'REAL':1})
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42, stratify=y
)
```

```
model = MultinomialNB()
model.fit(X_train, y_train)

print("Model parameters:", model.get_params())
```

```
Model parameters: {'alpha': 1.0, 'class_prior': None, 'fit_prior': True, 'force_alpha': True}
```

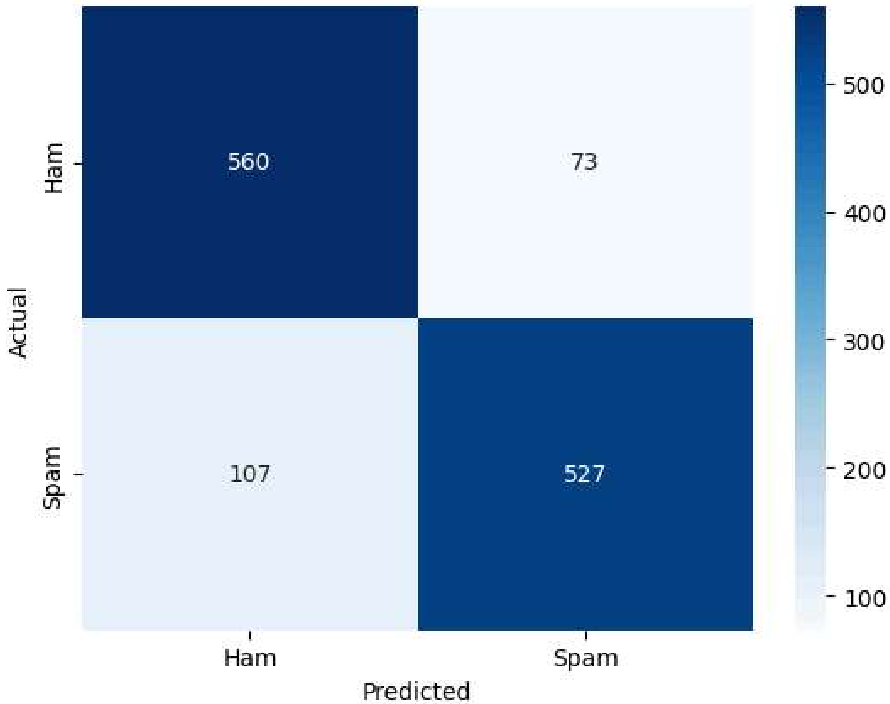
```
y_pred = model.predict(X_test)

print("Accuracy:", accuracy_score(y_test, y_pred))
print("Precision:", precision_score(y_test, y_pred))
print("Recall:", recall_score(y_test, y_pred))
print("F1-score:", f1_score(y_test, y_pred))

# Confusion Matrix
cm = confusion_matrix(y_test, y_pred)
sns.heatmap(cm, annot=True, fmt='d', cmap='Blues', xticklabels=['Ham','Spam'], yticklabels=['Ham'
plt.xlabel("Predicted")
plt.ylabel("Actual")
plt.show()

print("\nClassification Report:\n", classification_report(y_test, y_pred))
```

```
Accuracy: 0.8579321231254933
Precision: 0.8783333333333333
Recall: 0.831230283911672
F1-score: 0.8541329011345219
```




```
Classification Report:
              precision    recall  f1-score   support

           0       0.84      0.88      0.86       633
           1       0.88      0.83      0.85       634

    accuracy                           0.86      1267
   macro avg       0.86      0.86      0.86      1267
weighted avg       0.86      0.86      0.86      1267
```